

Scam Signatures: Discovering Fraud Families in Online Job Postings

Project 1, Milestone 3: Final White Paper

Author: Timothy Hollis
Bellevue University
DSC680 – Applied Data Science
Professor Xu Ashton
Date: October 4, 2026

Business Problem and Research Questions

Job scams are a growing source of consumer harm. The Federal Trade Commission reported more than \$220 million in job scam losses in the first half of 2024 alone (Federal Trade Commission, 2024). Existing detection tools, including a classifier and a fine-tuned language model I built on this dataset in earlier courses, answer only one question: is this posting fraudulent? A verdict without an explanation is hard for a job seeker to act on and hard for a trust and safety team to audit. This project asks instead whether fraudulent postings organize into a small, explainable set of families based on their language alone.

Three research questions follow. First, when postings are grouped by textual similarity with the fraud label withheld, do fraudulent postings concentrate in a small number of clusters or scatter among legitimate ones? Second, how many distinct signatures exist, and what distinguishes each one? Third, do postings written in 2025 and 2026 fall into those signatures or form new groups?

My hypotheses were that fraud would concentrate, that a minority of clusters would hold a majority of it, and that signatures would be defined as much by what a posting omits as by what it says. All three are answered here.

Background and History

A job posting is a document a stranger is invited to trust, and applying requires handing over exactly the information an identity thief wants. Unlike phishing email, a fraudulent posting is published openly on a platform with a commercial interest in volume, so detection has to happen at scale rather than through individual reporting.

The Employment Scam Aegean Dataset (EMSCAD) was assembled to support research on this problem (Vidros et al., 2017). It contains real postings collected through a recruiting platform between 2012 and 2014, annotated by that platform's own specialists. Most published work treats it as supervised classification, which produces accurate verdicts and no explanations. Reframing it as unsupervised discovery asks a different question: not whether a model can separate fraud from legitimate postings, but whether fraud has recognizable internal structure a human reviewer could learn to see.

Data Explanation

The dataset contains 17,880 postings, of which 866 (4.84%) are annotated fraudulent. Each posting carries five free text fields, seven categorical fields, three binary flags, and a salary range. I worked from the public Kaggle mirror (Bansal, 2020), which preserves the original 17,880 rows and 18 columns.

Preparation ran in five reproducible steps. I joined the five text fields into one document per posting and stripped residual HTML. The dataset replaces URLs, emails, and phone numbers with placeholder tokens; I counted those as a structural feature before normalizing them to plain words, since a direct contact instruction is informative while a random hash is noise. I recorded field presence flags before any missing values were filled, because a missing company profile is a signal rather than a defect. I then removed one posting with no description, one document under five words, and 2,006 exact duplicates, leaving 15,872 postings at a 4.48% fraud rate.

Deduplication removed a real signal, since fraudulent postings were duplicated at 22.3% against 15.0% for legitimate ones. Density-based clustering is unusually vulnerable to duplicates, because twenty identical copies create a dense region that is one posting counted twenty times rather than a pattern. I therefore recorded each posting's copy count as a feature before collapsing duplicates.

Key fields are summarized below.

Field	Type	Description
title, company_profile, description, requirements, benefits	Text	The five source text fields, joined into one document for modeling

Field	Type	Description
telecommuting, has_company_logo, has_questions	Binary	Platform flags for remote work, employer logo, and screening questions
employment_type, required_experience, required_education, industry, function, location, department	Categorical	Used to describe clusters, not to form them
fraudulent	Binary	Target label, withheld during clustering and revealed only for evaluation
document, doc_length	Derived	Cleaned combined text and its word count
n_copies	Derived	Number of times the identical document appeared before deduplication
n_contact_tokens	Derived	Count of URL, email, and phone placeholders in the source text
has_title, has_company_profile, has_description, has_requirements, has_benefits, has_salary	Derived	Field presence flags captured before any imputation
cluster, cluster_tfidf, archetype	Derived	Cluster assignments under each representation, and the archetype label

Methods

The pipeline represents each posting as a vector, reduces the dimensionality, clusters, and interprets. For representation I used sentence embeddings from all-MiniLM-L6-v2 (Reimers & Gurevych, 2019), which place documents with similar meaning near one another, chosen on the assumption that scam postings paraphrase a shared pitch and that semantic similarity would outperform exact word matching. The model has a 256 token limit, which truncated 83.9% of documents unevenly: because fields are concatenated in a fixed order, the description was never seen for 12.2% of postings, requirements for 67.8%, and benefits for 81.3%. Since compensation claims live in benefits, this was a defect rather than a limitation, so I split each document into 180 word chunks, embedded each, and averaged them.

I applied UMAP (McInnes et al., 2018), which compresses many dimensions into few while preserving local neighborhoods, twice: five components with a minimum distance of zero as the clustering space, and two components for visualization. Compressing to two dimensions discards structure that density-based clustering depends on, so one reduction serving both purposes would have cost real information.

I used HDBSCAN (McInnes et al., 2017) as the primary clustering method because it finds groups by density, determines the number of clusters from the data and can assign ambiguous points to noise rather than forcing them into a group. The minimum cluster size of 25 was selected on internal evidence before the labels were revealed. A sweep showed silhouette scores (Rousseeuw, 1987) holding steady through 40 and then falling, with the largest cluster jumping from 335 to 750 members over the same interval, indicating that distinct groups merge past that point. I chose 25 over 40 on an asymmetry of errors: a family of thirty postings is discarded as noise at 40 and never appears in the results, whereas a family split across two clusters at 25 announces itself on reading.

Two baselines tested those choices independently: K-means on the same reduced space tested the algorithm, and TF-IDF with truncated SVD, which scores documents on their distinguishing words rather than on meaning, substituted for the sentence encoder tested the representation, everything else held constant. All methods were scored on one yardstick, the share of fraud captured by clusters running at three times base rate divided by the share of corpus they occupy. That measure is fair to methods that leave points unassigned and to methods that do not, and it matches the practical question a reviewer faces.

Validation ran in both directions: internally through silhouette scores and stability across three seeds, and externally by revealing the withheld label only after the clustering was fixed. I deliberately did not report adjusted Rand index against the label, because comparing a 173 way partition to a two way one penalizes exactly the splitting the method was asked to perform and would produce a near zero score that misrepresents the result.

Analysis

Stability

Across UMAP seeds of 42, 7, and 2026 the pipeline produced 173, 172, and 190 clusters at 31.4%, 29.1%, and 31.0% noise. Pairwise agreement on the roughly 9,800 postings clustered in every run gave adjusted Rand indices of 0.898 to 0.937 and normalized mutual information above 0.97 (Hubert & Arabie, 1985). The structure is in the data rather than in any single initialization, though about one thousand boundary postings move between clustered and noise depending on the seed.

Postings fall into one dense crowd ringed by small, tight satellites

15,872 postings in two dimensional embedding space, fraud labels withheld

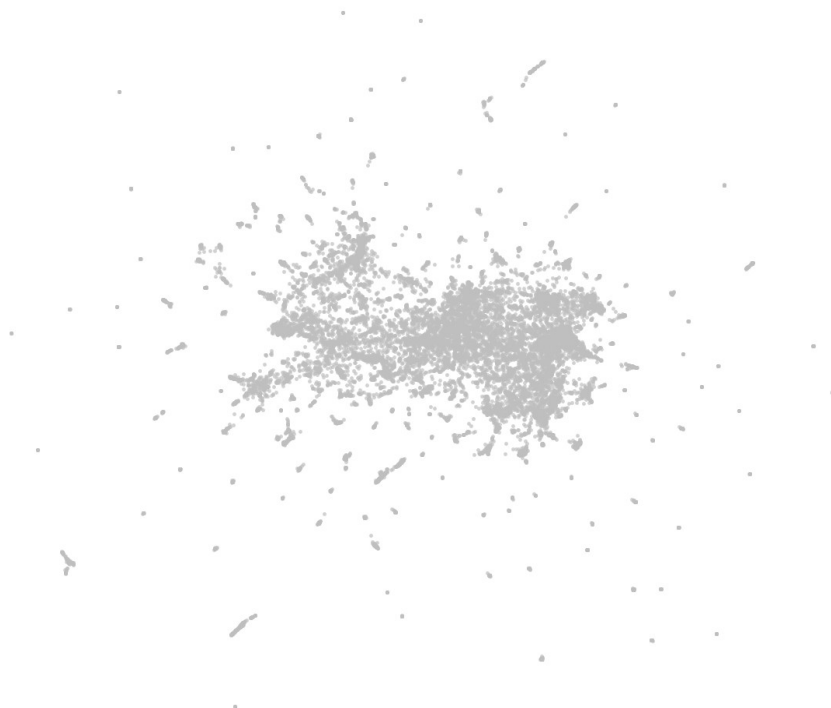


Figure 1. Postings in two dimensional embedding space, labels withheld. A dense central mass of ordinary postings is surrounded by dozens of small, well separated satellite groups.

Research question one: concentration

Fifteen clusters ran at three times base rate or higher, holding 302 of 711 fraudulent postings (42.5% of all fraud) within 910 postings, or 5.7% of the corpus. The two densest ran at 96.8% and 90.0% fraudulent.

The more informative number is the noise group. Postings HDBSCAN declined to classify carried 4.01% fraud, below the 4.48% corpus rate. Had fraud been mostly improvised, the unclustered middle would have carried it at or above base rate. It carried less, which is direct evidence that fraud preferentially occupies dense, template shaped regions.

The qualification is that 57% of fraudulent postings were not in a high lift cluster under this representation. The answer to research question one is therefore both: a substantial minority of fraud is

organized into recognizable families, and the rest is either genuinely one off or camouflaged inside ordinary groups.

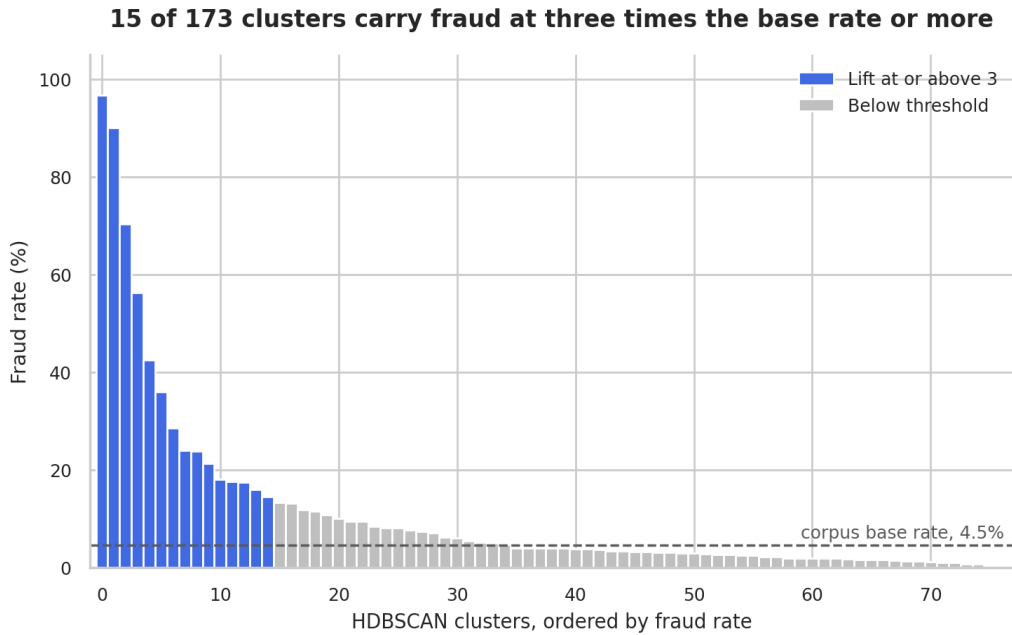


Figure 2. Fraud rate by cluster against the corpus base rate, showing the small number of clusters carrying high lift.

Comparison of methods

The baseline comparison reversed my starting assumption.

Method	Fraud captured	Corpus occupied	Unassigned	Concentration
HDBSCAN on embeddings	42.5%	5.7%	31.4%	7.4
K-means (k = 173) on embeddings	55.8%	8.2%	0.0%	6.8
HDBSCAN on TF-IDF	61.0%	6.6%	27.3%	9.3
K-means (k = 173) on TF-IDF	67.7%	9.7%	0.0%	7.0

TF-IDF outperformed sentence embeddings on both recall and precision, reversing my proposal's premise that semantic similarity would matter because scam postings paraphrase a shared pitch. These families do not paraphrase; they copy and paste. Lexical overlap is therefore the stronger signal, and an uncased 384 dimensional representation actively discards the brand names, dollar figures, and misspellings that identify a template.

The K-means runs also showed that the silhouette based selection of k my proposal specified does not select in this data. Silhouette rose monotonically from 0.324 at k = 10 to 0.502 at k = 173 with no elbow, because the data is one large mass plus many small satellites, so every additional centroid finds another satellite. With K-means, nothing tells you how many clusters to request.

The two representations proved complementary rather than redundant. Of 711 fraudulent postings, 226 were captured by both, 76 by embeddings only, 208 by TF-IDF only, and 201 by neither. Their union captured 510 postings, 71.7% of all fraud, within 9.8% of the corpus. They answer different questions: lexical similarity finds families organized by the claimed identity of the poster, semantic similarity finds families organized by the nature of the offer.

A supervised check confirms the representation finding by an independent route. Holding the classifier constant and varying only the representation, logistic regression reaches 0.855 average precision on raw TF-IDF against 0.427 on sentence embeddings, exactly double. The result is not an artifact of dimensionality, since

embeddings at 384 dimensions lose to TF-IDF compressed to 100 components, which reaches 0.568. A decision tree scores below both on every representation and ranks them differently, which reflects how poorly axis aligned splits suit high dimensional sparse text rather than anything about the representations themselves. Appendix A4 gives the full comparison. Labels are used there only to rank representations and appear nowhere in the clustering pipeline.

That check also measured how much of a supervised score is memorization rather than generalization. Trained with part of the Ghost Agency campaign in view, the model flags 100% of the held out remainder. Trained with the campaign removed entirely, it flags 14.1% of it, while performance on non campaign postings is unchanged at 0.823 against 0.825. A classifier is therefore close to blind to campaigns it has not already been shown, and a new campaign is by definition one it has not been shown. That gap is what unsupervised discovery exists to fill, and it is the strongest evidence in this paper for the sequence the recommendations describe.

One further consequence deserves note. Compressing TF-IDF to 100 components costs roughly a third of the available signal, and the clustering pipeline ran on that reduced space. TF-IDF therefore won the clustering comparison while working with a degraded version of itself, which suggests clustering the sparse representation directly is worth testing.

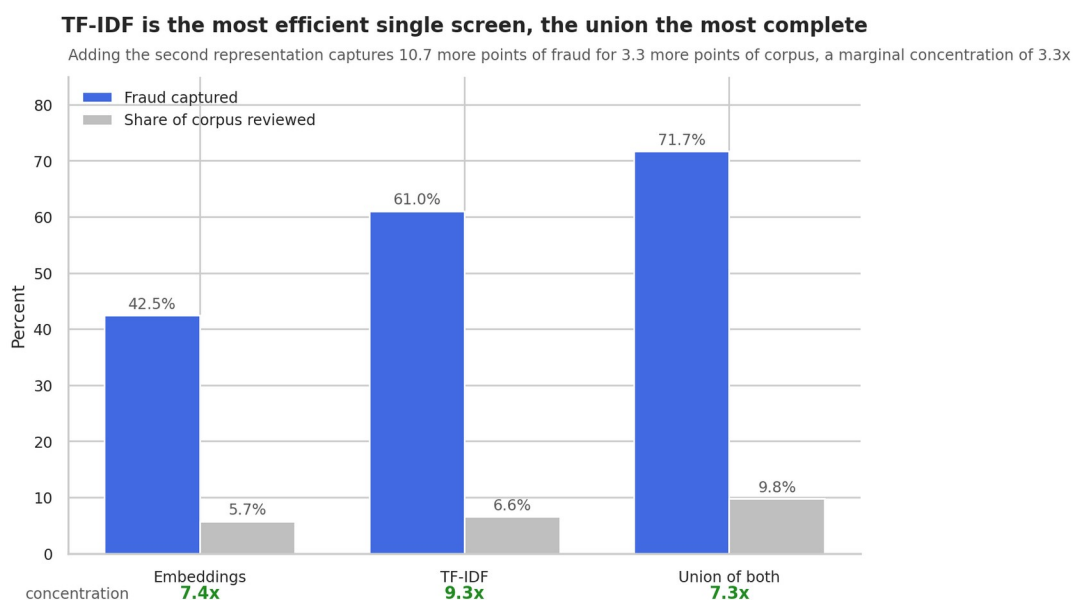


Figure 3. Fraud captured by each representation and by their union, against the share of the corpus reviewed.

Research question two: the archetypes

Reading the fraud dense clusters under both representations produced four recognizable archetypes, together accounting for 504 postings, 3.2% of the corpus, containing 47.5% of all fraud.

Archetype	Postings	Fraud rate	Company profile	Logo	Screening questions	Median words
The Bait Ad	71	92.9%	0.0%	4.2%	5.6%	81
The Ghost Agency	99	100.0%	100.0%	100.0%	89.9%	571
The Borrowed Brand	51	100.0%	58.8%	58.8%	0.0%	514
The Common Form	283	43.1%	36.8%	40.6%	40.3%	208
Corpus baseline	15,872	4.5%	80.7%	80.0%	49.1%	366

The Bait Ad targets the job seeker directly and is recognizable entirely by absence. Not one carries a company profile, screening questions appear on 5.6% against 49.1% for the corpus, and at a median of 81 words

they run a fifth the length of a typical posting while flagging telecommuting at fifteen times base rate. The offer is remote work, no experience, immediate start, with guaranteed wages spanning an implausible range. This is the one archetype a simple structural filter catches reliably.

The Ghost Agency is the most consequential finding here. A single operator advertised 60 distinct professional roles across 22 locations under at least two brand names, every posting opening with the same signature paragraph: 99 postings, all fraudulent, 13.9% of all fraud in the dataset. No two documents are identical, because the job specific body varies enough to defeat duplicate detection while the opening paragraph is reused verbatim, and shared misspellings confirm single authorship. Structurally these postings are flawless, carrying a company profile and logo on 100% of listings and screening questions on 89.9%, better presented than the average legitimate posting. Every absence based heuristic fails on them completely.

The Borrowed Brand copies a real multinational's corporate description verbatim, including a duplicated section header, and attaches it to invented roles across 32 job titles. All 51 postings are fraudulent. The stolen text supplies credibility, so profile and logo appear near corpus norms. One tell survives: screening questions appear on exactly 0.0%, because nobody adds an application step to a vacancy that does not exist.

The Common Form is the highest volume archetype and the least conclusive: stock job descriptions from free template libraries, generic administrative listings, and short informal advertisements, spanning 200 job titles at a 43.1% fraud rate. Scammers and low effort legitimate employers reach for the same forms because they are free and look adequate, and every cluster within this archetype contains real employers. It indicates how much scrutiny a posting warrants, never a verdict.

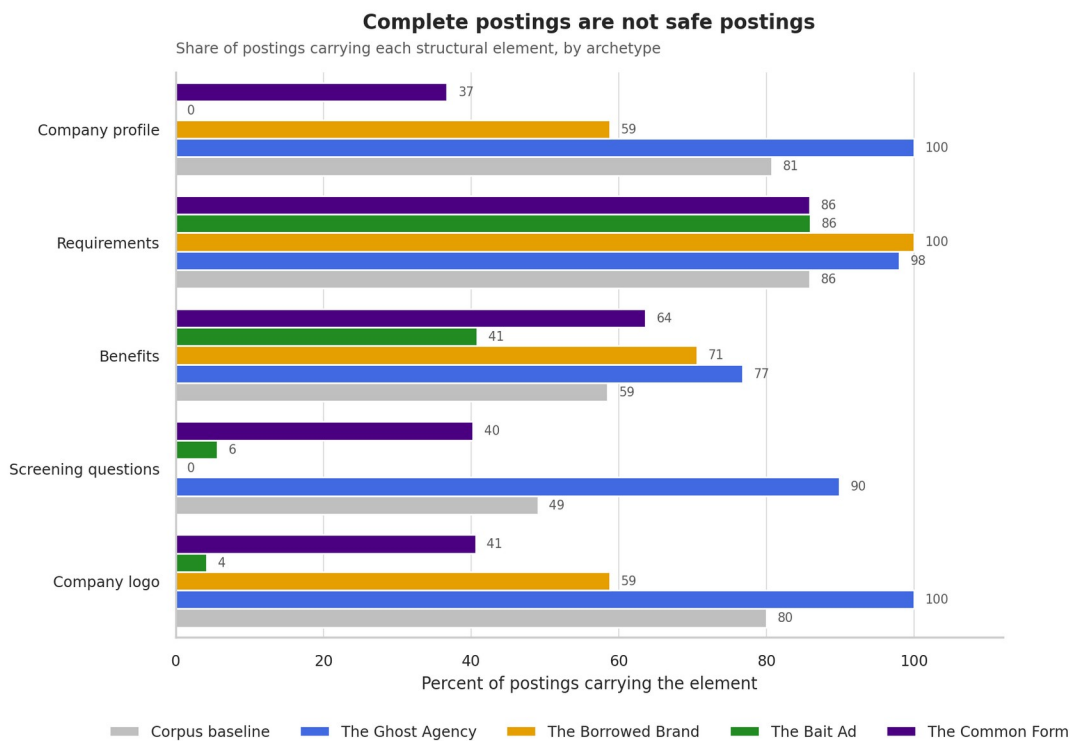


Figure 4. Structural fingerprint of each archetype against the corpus baseline, showing that completeness does not indicate legitimacy.

Together the archetypes reverse an intuition underlying most heuristic fraud detection. The crudest fraud is easiest to catch and the most professional is hardest, and the structural completeness that reads as legitimacy is precisely what the largest operator in this dataset invested in.

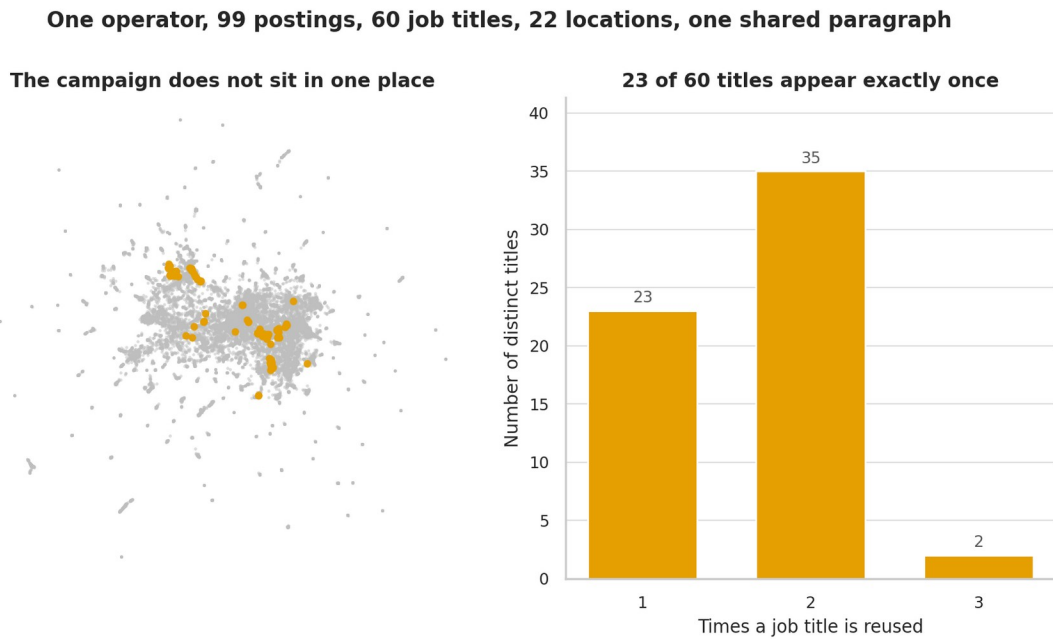


Figure 5. The Ghost Agency campaign: 99 postings, 60 job titles, 22 locations, all sharing one signature paragraph.

Research question three: persistence in current postings

The first two questions were answered inside the historical corpus. The third asks whether the shapes found there describe postings written today, and it can be answered without any labels at all. I collected 165 job postings in 2026 from three public job board interfaces, masking each employer name at collection before anything was written to disk, then scored them against the fitted vectorizer and the archetype centroids. Nothing was refitted, so current text is judged entirely by historical vocabulary.

The signatures do not carry forward. Ordinary historical postings sit at a median similarity of 0.100 to their nearest archetype and archetype members at 0.289. The modern sample sits at 0.133, roughly 18% of the distance from an ordinary posting to an archetype member. Document length does not explain that gap. Matched against the 6,414 historical postings of comparable length, the baseline rises only to 0.110 while the modern sample rises to 0.145, leaving the difference essentially unchanged.

Two further measurements say the same thing from different directions. Only 56.2% of the words in a current posting are recognized by a vocabulary fitted in 2014, so close to half of modern posting language is invisible to the model. And 96.4% of modern postings land nearest the Common Form, the one archetype this paper declines to treat as evidence, which is the signal that the discriminating archetypes have stopped discriminating. Neither the Bait Ad nor the Borrowed Brand attracted a single modern posting, which is worth stating as a precision result: the method raised no fraud shaped alarm against any of the 165 legitimate listings it was shown.

The sample is small and drawn from curated boards, so it is almost entirely legitimate. It therefore tests whether these shapes persist in current language, not whether they would catch current fraud. Answering the second question would require a labeled modern corpus, which is the work this result makes necessary.

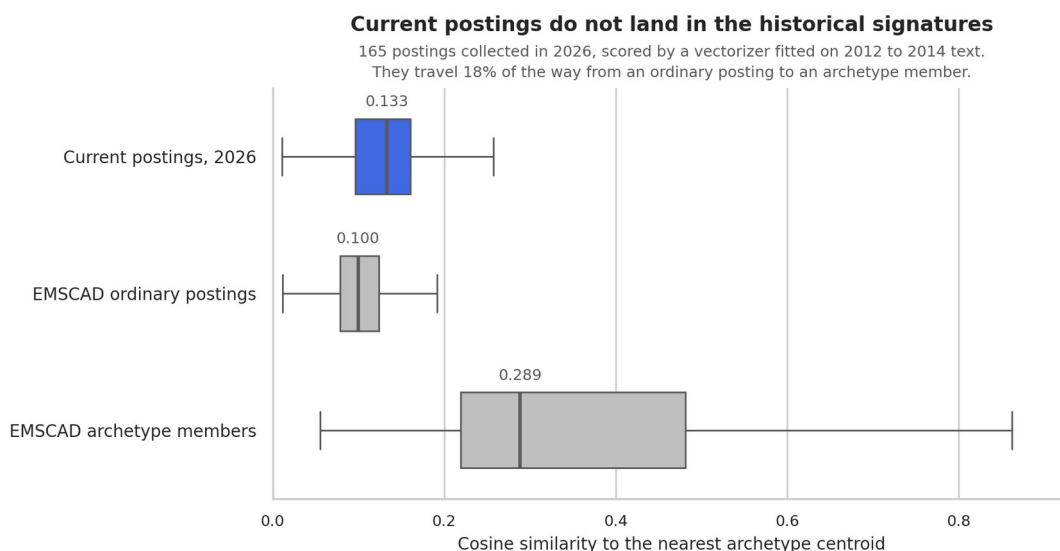


Figure 6. Similarity to the nearest archetype for current postings, against historical postings that belong to an archetype and those that do not.

Conclusion

Fraudulent job postings both concentrate and scatter. Roughly seventy percent of fraud can be surfaced by grouping postings on language alone, within under ten percent of the corpus, and the remaining thirty percent is not reachable by clustering at all. Fraud organizes along two independent axes, the offer being made and the identity being claimed, and a method capturing one will miss much of the other. Those findings describe the corpus they came from. Scored against postings written in 2026, the same archetypes travel only about a fifth of the way from an ordinary posting to a recognizable one, so the shapes are real but dated.

For a trust and safety team the practical result is that reviewing roughly one posting in ten, selected by these methods, surfaces seven of every ten scams. For a job seeker the result is a vocabulary: four recognizable shapes a fraudulent posting takes, three of which can be described without any statistical training.

Assumptions

I assumed the platform's fraud annotations approximate ground truth closely enough to serve as an evaluation reference, that patterns present in 2012 to 2014 postings retain some relevance to current fraud, that the five text fields carry the signal of interest, and that cosine distance on normalized embeddings and Euclidean distance on the reduced space are appropriate similarity measures for this data.

Limitations

The most serious limitation concerns the labels. Within one recruiter's postings I found byte identical company descriptions where the dataset labels one posting fraudulent and two others legitimate. If the label distinguishes documents that are textually indistinguishable, it encodes information outside the text, most plausibly complaints received or account action taken after publication. No text based method can recover that, and it places a ceiling on what any result here, mine or anyone's, can claim.

Approximately 28% of fraudulent postings fall outside every cluster either method identified. A single operator accounts for 13.9% of all fraud, so the concentration result partly measures one actor being prolific rather than many distinct families existing. The embedding model is uncased and therefore blind to capitalization and punctuation irregularities a human reader treats as strong signals. Mean pooling weights long fields more heavily than short ones. Finally, the dataset is more than a decade old, and modern scam categories such as the paid coaching call are absent entirely.

Challenges

Three challenges shaped the work. Truncation was discovered only after the first embedding run, when a diagnostic showed the benefits field was never seen for 81.3% of postings; rebuilding with chunking cost a pipeline stage but corrected a defect that would have invalidated the results. The comparison baselines reversed my central methodological assumption, requiring the methods argument to be rewritten around the finding rather than the plan. And the minimum cluster size produced a family split across two clusters, which I recovered by testing shared phrases, confirming one operator at 100% precision across four template phrases.

Future Uses and Additional Applications

The immediate extension is a labeled modern corpus, which the drift result makes necessary: this project can say the historical shapes no longer describe current postings, but not yet what has replaced them. Typography features such as capitalization consistency and misspelling rates would capture signal the embedding model structurally cannot see, and embedding each field separately would prevent long marketing text from outweighing short compensation claims. Clustering the sparse representation directly, rather than the reduced one, is worth testing given how much signal the reduction costs. The method is domain general and would apply to rental listings, marketplace advertisements, or any venue where a fraudulent offer must imitate a legitimate one.

Recommendations

First, run both representations rather than choosing between them. They find different fraud, and the union captures 71.7% of scams within 9.8% of the corpus, which neither achieves alone.

Second, use clustering to discover campaigns and then extract their distinctive phrases as exact match rules. In this dataset four template phrases identified 99 postings at 100% precision, including three that the clustering itself missed. Discovery is expensive and matching is nearly free, so the sequence matters.

Third, treat the archetype as a scrutiny level rather than a verdict. Three of the four are strong fraud indicators; the fourth contains real employers and must never be used to act against a poster automatically.

Fourth, do not treat structural completeness as evidence of legitimacy. The largest fraud operator in this dataset scored higher on company profile, logo, and screening questions than the average genuine employer. Employer verification, not posting quality, is the only defense against the Borrowed Brand and the Ghost Agency.

Implementation Plan

The archetype work was folded into Verify This Job, the fine tuned classifier application this project extends. The original plan was to ship the fitted TF-IDF vectorizer and a matrix of archetype centroids and report each posting nearest archetype. The third research question ruled that out. With 96.4% of current postings landing nearest the Common Form, a centroid matching signal would report the one archetype this paper refuses to treat as evidence, for nearly every posting a real user submits, with the outward appearance of precision.

What shipped instead carries no model at all. The two findings that transfer are exact campaign phrases and structural absence, and both are plain text operations. A standalone module holds the four Ghost Agency phrases as exact match rules and detects the Bait Ad shape from brevity combined with the absence of any employer description and any stated requirement. It depends on nothing beyond the Python standard library, which also means it returns an answer while the language model is still loading, and it carries forward unchanged if the application is later rebuilt on a different framework.

Precedence follows the evidence. A campaign phrase match escalates the application verdict on its own, overriding a legitimate classification, because those four phrases identified 99 postings with no false positives and because the supervised check showed the classifier catches only 14.1% of a campaign it has not seen. A structural match raises a caution and never more, because short legitimate postings can carry that shape. The Borrowed Brand and the Common Form are not screened for at all, the first because identifying it needs corpus comparison and platform metadata the application does not have, the second because it is not evidence.

One signal was added during integration rather than discovered by the analysis. The Bait Ad description notes guaranteed wages spanning an implausible range but never operationalized it, so the module flags any advertised range whose upper figure is at least three times its lower. Measured against the corpus afterward, 42 postings match, of which 21.4% are fraudulent against the 4.48% baseline, a lift of 4.8. The sample is small enough that the 95% interval runs from 11.7% to 35.9%, and it never fires alone, since the Bait Ad shape also requires brevity and both absences.

Monitoring for drift requires no labels, which matters because a deployment has none. Three quantities suffice. The distribution of similarity to the nearest archetype says whether incoming postings still resemble the shapes. The share of posting vocabulary the fitted model recognizes says whether it can still read current language. The share of postings whose nearest archetype is the Common Form says whether the discriminating archetypes still discriminate. All three moved measurably in this project before any label was consulted, reaching 0.133, 56.2% and 96.4% respectively. A production version would alert when coverage falls below a set floor or when median similarity converges on the baseline, and either alert means the centroids need refitting rather than that fraud has stopped.

For a job board rather than a consumer tool, the same components would run as a nightly batch, with high lift archetype matches routed to human review and confirmed campaigns converted into phrase rules applied at submission time.

Ethical Assessment

Seven concerns apply to this project. Each is identified, assessed for likelihood and impact, and paired with the mitigation applied.

Concern	Likelihood	Impact	Mitigation applied
Harm to legitimate employers whose postings fall inside fraud dense groups	High	Moderate to high	Report fraud rates per cluster rather than labeling clusters as fraudulent; present archetypes as scrutiny levels, not verdicts; no employer named in any published output
Label validity, given annotations assigned by platform staff more than a decade ago	High	Moderate	Treat labels as a reference point rather than truth; report the case of identical text carrying opposite labels; state the resulting ceiling explicitly
Privacy of individuals and organizations named in the postings	Moderate	Moderate	All organization and individual names masked in every displayed output through a redaction layer applied before printing
Potential for misuse, since describing what makes a posting recognizable also describes how to avoid recognition	Low	Moderate	Signatures described at the level of themes and structural patterns; no publishable word list or template text reproduced
Fairness across industries and regions, if scam heavy sectors become proxies for suspicion	Moderate	Moderate	Industry and location composition reported alongside fraud rates; archetypes defined by behavior rather than sector
Researcher bias in interpretation, since I chose which clusters to read and how to name them	Moderate	Moderate	Representative members selected by membership confidence or random sample with a fixed seed rather than by hand; naming grounded in printed evidence
Data sourcing and terms of use	Low	Low	Public research dataset used under its stated license; no scraping; every transformation documented in the notebook

Changes made to the data. I removed 2,008 rows: one posting with no description, one document under five words, and 2,006 exact duplicates. I derived ten features and normalized anonymization placeholders to

plain words. Nothing was imputed and no outliers removed. Every filtering decision is stated with its count in the notebook; the five word minimum was set only after measuring that very short postings are fraud enriched, so it removed degenerate rows without discarding signal. Nothing was filtered silently.

Legal standing, presentation risk, and sourcing. The dataset is published for research use, contains no personal information beyond what employers chose to publish, and no protected attributes are used as features. A production deployment acting against real employers would raise fair notice questions a research analysis does not, which is why the recommendations stop at routing for human review. Two transformations carry risk: deduplication removed a genuine fraud signal, mitigated by recording copy counts first, and mean pooling underweights short fields. Presenting fraud rates by cluster invites a reader to treat membership as a verdict, which the Common Form archetype makes unsafe, so that caution is repeated in the Conclusion and the Recommendations. The identity masking applied here reduces amplification of names in published results but cannot make a public research dataset private, and the redaction list itself remains visible in the notebook source.

Code and Reproducibility

The complete analysis is contained in a single notebook, `scam_signatures.ipynb`, which runs top to bottom and reproduces every figure and table in this paper. The notebook and the six figures are available at <https://github.com/TTHolliis/DataSciencePortfolio/tree/main/projects/ScamSignatures>. The screening module that shipped from this work, `archetype_screen.py`, lives with the application it was written for at <https://github.com/TTHolliis/VerifyThisJob> and carries a self test that verifies it still recognizes a known campaign posting. Embeddings are cached to avoid recomputation and all random seeds are fixed. The source dataset is not redistributed; the notebook locates it or prompts for it.

References

- Bansal, S. (2020). *Real or fake: Fake job posting prediction* [Data set]. Kaggle.
<https://www.kaggle.com/datasets/shivamb/real-or-fake-fake-jobposting-prediction>
- Federal Trade Commission. (2024, December 12). *New FTC data show skyrocketing consumer reports about game-like online job scams* [Press release].
<https://www.ftc.gov/news-events/news/press-releases/2024/12/new-ftc-data-show-skyrocketing-consumer-reports-about-game-online-job-scams>
- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218.
<https://doi.org/10.1007/BF01908075>
- McInnes, L., Healy, J., & Astels, S. (2017). hdbscan: Hierarchical density based clustering. *Journal of Open Source Software*, 2(11), 205. <https://doi.org/10.21105/joss.00205>
- McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. arXiv. <https://doi.org/10.48550/arXiv.1802.03426>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing* (pp. 3982–3992). Association for Computational Linguistics. <https://doi.org/10.18653/v1/D19-1410>
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Vidros, S., Kolias, C., Kambourakis, G., & Akoglu, L. (2017). Automatic detection of online recruitment frauds: Characteristics, methods, and a public dataset. *Future Internet*, 9(1), 6.
<https://doi.org/10.3390/fi9010006>

Appendix A: Supporting Documentation

A1. Archetype to cluster mapping

Archetypes were assigned in precedence order so that a posting appearing under both representations receives a single label. The Ghost Agency was assigned first because its membership is phrase verified at 100% precision, followed by the Borrowed Brand, the Bait Ad, and the Common Form.

Archetype	Source representation	Cluster identifiers	Assignment basis
The Ghost Agency	TF-IDF	37, 41, plus three outside	Exact match on four template phrases
The Borrowed Brand	TF-IDF	2	Cluster membership
The Bait Ad	Embeddings	66, 35	Cluster membership
The Common Form	Both	170, 171, 138, 139, 67	Cluster membership

A2. Template phrase evidence for the Ghost Agency

Four phrases drawn from the two clusters were matched across the full corpus. Every posting containing any of them is annotated fraudulent, and the matches span both clusters, confirming a single operator behind two brand identities.

Phrase (abbreviated)	Postings	Fraud rate	Clusters spanned
"leveraging your career"	78	100%	37, 41, and three others
"redesigned the recruiting wheel"	35	100%	37
"have partnered up in an effort to streamline"	43	100%	37, 41, and three others
"Represented candidates"	56	100%	37, 41
Union of all four	99	100%	37, 41, and three others

A3. HDBSCAN parameter sweep

The minimum cluster size was chosen from this sweep before the labels were revealed. Silhouette scores are computed on the reduced space and are optimistic by construction, since dimensionality reduction increases separation; they are used here only to compare settings against one another.

Minimum cluster size	Clusters	Noise	Silhouette	Largest cluster
10	411	28.3%	0.701	269
15	280	28.4%	0.683	327
25 (selected)	173	31.4%	0.651	327
40	108	37.5%	0.629	335
60	64	36.1%	0.530	750
100	36	43.8%	0.539	769

A4. Supervised reference check

Representations were compared by holding the classifier constant and varying only the input. Stratified 75/25 split, fraud prevalence 4.48%. Accuracy is not reported because a model predicting legitimate for every posting would score 95.5% on it. Labels were used here only to rank representations and were withheld from the clustering throughout.

Representation	Model	Precision	Recall	F1	PR AUC
Sentence embeddings, 384d	Logistic regression	0.210	0.798	0.333	0.427
Sentence embeddings, 384d	Decision tree	0.210	0.427	0.281	0.138
TF-IDF then SVD, 100d	Logistic regression	0.198	0.888	0.324	0.568
TF-IDF then SVD, 100d	Decision tree	0.241	0.775	0.368	0.268
TF-IDF raw, sparse	Logistic regression	0.740	0.831	0.783	0.855
TF-IDF raw, sparse	Decision tree	0.150	0.820	0.254	0.162

Campaign memorization was measured separately. With part of the Ghost Agency campaign in training, the model flags 100% of the held out remainder. With the campaign removed from training entirely, it flags 14.1% of its 99 postings, while PR AUC on the same non campaign test set is unchanged at 0.823 against 0.825.

Ten Questions an Audience Would Ask

These are the ten questions an informed audience would put to this work, each answered in full. Figures, tables and appendices referenced in the answers appear earlier in this paper.

1. This dataset was used in two of your previous courses. What is genuinely new here, and how do you avoid simply re-reporting earlier results?

The earlier work was supervised. I built a classifier in one course and fine-tuned a language model in another, and both answered a single question: is this posting fraudulent. This project withholds the label entirely during modeling and asks a structural question instead, which is whether fraud has internal organization discoverable from language alone. The label is revealed only after the clustering is fixed, and only to score what was found.

Nothing from the earlier models is reused. No feature set, no threshold, no trained weights. The outputs also differ in kind. The earlier work produced a verdict per posting, which a reviewer cannot audit and a job seeker cannot learn from. This produces four named shapes with structural fingerprints a person can recognize without any statistical training.

The clearest sign that this is not a re-report is that the headline finding contradicts what the supervised work would have predicted. Structural completeness reads as legitimacy, and the largest fraud operator in the dataset presents better than the average genuine employer.

The relationship between the two projects now runs in one direction. The findings from this work were folded back into that earlier application as a screening module, while nothing from the earlier models came the other way. A re-report would have had nothing to contribute.

2. How do you know your clusters are capturing fraud patterns rather than industries? Several of your high lift clusters look like occupational categories.

For one of the four archetypes, I do not, and that is the honest core of the Common Form finding. It spans 200 job titles at a 43.1% fraud rate and every cluster within it contains real employers. That is exactly this objection made concrete, which is why I report it as a scrutiny level and never as a verdict.

The other three are defined by properties that cut across occupation rather than following it. The Ghost Agency spans 60 distinct professional roles across 22 locations under two brand names, so occupation is precisely what it crosses; its membership is confirmed by four exact template phrases matching 99 postings at 100% precision, with shared misspellings indicating single authorship. The Borrowed Brand spans 32 job titles and is defined by corporate text copied verbatim from a real multinational. The Bait Ad is defined by absence, with zero company profiles at a median of 81 words, which is a property of the document rather than of a sector.

The general answer is the structural fingerprint. The archetypes differ from the corpus baseline on company profile, logo and screening question rates by 40 to 80 percentage points, and none of those fields has anything to do with what job is being advertised.

3. The data is from 2012 to 2014. Why should a trust and safety team today act on patterns found in decade old postings?

Largely they should not, and rather than argue the point I tested it. I collected 165 job postings in 2026 and scored them against the fitted archetype centroids. Ordinary historical postings sit at a median similarity of 0.100 and archetype members at 0.289. The modern sample sits at 0.133, roughly 18% of the distance from an ordinary posting to an archetype member, and that result holds after matching on document length against 6,414 comparable historical postings. The signatures do not transfer.

Two things do transfer. The method is dataset independent: cluster on language, read the dense groups, extract their distinctive phrases as exact match rules. And the structural insight is about how reviewers reason rather than about 2014 vocabulary, so the finding that completeness is not evidence of legitimacy survives the drift.

What a team should take from this is the recipe and the caution, then refit on their own current data. Acting on these specific centroids today would be a mistake, and I would rather say so than sell a result the evidence does not support.

4. A single operator accounts for 13.9% of all fraud in your dataset. Does that mean your concentration finding is really a finding about one scammer?

Partly yes, and I state it as a limitation rather than hiding it. But the concentration result does not rest on that operator. Fifteen clusters running at three times base rate hold 302 of 711 fraudulent postings. The Ghost Agency contributes 99 of those, so removing it entirely still leaves roughly 200 fraudulent postings concentrated inside a small share of the corpus.

The stronger evidence is independent of any single actor. Postings the clustering declined to assign carry 4.01% fraud against a 4.48% corpus rate. Had fraud been mostly improvised one-offs, that unclustered middle would have carried fraud at or above the base rate. It carries less, which is a statement about the whole corpus rather than about one campaign.

There is also a substantive reading worth offering. Fraud at scale is committed by a small number of prolific operators. A concentration finding that partly measures one prolific actor may be describing the domain accurately rather than suffering from an artifact.

5. The literature generally favors sentence embeddings over TF-IDF for text clustering. Why did the opposite hold here, and would it hold on a different corpus?

Because the premise that made embeddings attractive turned out to be false for this data. I assumed scam postings paraphrase a shared pitch, which would favor semantic similarity. They do not paraphrase, they copy and paste. Once the task is finding reused text, lexical overlap is the signal, and an uncased 384 dimensional representation actively discards the brand names, dollar figures and misspellings that identify a template.

The margin was not small. Density based clustering on TF-IDF captured 61.0% of all fraud inside 6.6% of the corpus, a concentration of 9.3, against 42.5% inside 5.7% and a concentration of 7.4 for sentence embeddings.

I would expect the same result wherever the target behavior is duplication, such as template spam, coordinated review fraud or listing fraud, and the opposite wherever the target is a concept expressed in varied language. The more useful finding is that the two are complementary rather than competing. They caught 226 of the same fraudulent postings, 76 and 208 respectively that the other missed, and 201 that neither found. Their union reaches 71.7% of all fraud within 9.8% of the corpus. Lexical similarity finds families organized by the identity the poster claims; semantic similarity finds families organized by the nature of the offer.

6. What happens to a legitimate employer whose posting lands in a cluster that is 90% fraudulent? What recourse would they have?

This is the concern I rate highest for likelihood in the ethical assessment, and it is not hypothetical. In one cluster running at 90% fraud, the two most representative members by membership confidence are real gas station cashier advertisements. Under any automated action, those two employers would be penalized for writing a short, plain job ad.

That case is why the recommendations stop at routing for human review, why fraud rates are reported per cluster rather than clusters being labeled fraudulent, and why the Common Form must never trigger action against a poster. It is also why no employer is named in any published output.

In a real deployment, recourse has to be designed before the signal is. An employer should be told that a review is happening rather than silently deranked, and an appeal should reach a person rather than the model that flagged them. Membership in a dense group is evidence about a neighborhood of documents, not evidence about a company.

7. In production there are no labels. How would you know whether the archetypes were still working, or whether the clusters had drifted?

The procedure I used for the third research question is exactly that monitor, and it requires no labels at all. For any deployment that carries the fitted centroids, rather than the phrase and structure rules that shipped in the application, score incoming postings against those centroids and watch two unlabeled quantities over time: the distribution of similarity to the nearest archetype, and the share of posting vocabulary the fitted vectorizer still recognizes.

Both moved in this project and both were detectable without a single label. Modern similarity sits near the ordinary posting baseline rather than near archetype members, and vocabulary coverage is 56.2%, meaning nearly half of current posting language is invisible to a model fitted in 2014. A production version would alert when coverage falls below a set floor, or when median similarity converges on the baseline, either of which says the centroids need refitting.

A third check costs nothing: the share of postings whose nearest archetype is the Common Form. It is 96.4% in the modern sample, which is the signal that the discriminating archetypes have stopped discriminating.

8. You chose a minimum cluster size of 25. How sensitive are your conclusions to that choice, and what would change at 15 or 40?

The full sweep is in the appendix, and the parameter behaves smoothly across the range that matters. At 15 the pipeline produces 280 clusters at 28.4% noise with a silhouette of 0.683. At 25 it produces 173 clusters at 31.4% and 0.651. At 40 it produces 108 clusters at 37.5% and 0.629. Structure does not appear or vanish across that range, it subdivides.

The choice between 25 and 40 was an asymmetry of errors rather than a metric. At 40, a family of thirty postings is discarded as noise and never appears in the results, which is an error I cannot see. At 25, a family split across two clusters announces itself when I read them, which is an error I can correct. That is not hypothetical: the Ghost Agency landed across two separate clusters and I recovered it by testing shared phrases.

Past 40 the picture changes qualitatively. Between 40 and 60 the largest cluster jumps from 335 to 750 members, which is distinct groups merging rather than refining, and the silhouette falls from 0.629 to 0.530. Conclusions are stable from 10 through 40 and degrade beyond it.

Seed stability matters more than the size parameter, and I would rather be judged on it. Across three reduction seeds the pipeline produced 173, 172 and 190 clusters with pairwise adjusted Rand indices of 0.898 to 0.937 on the postings clustered in every run.

9. If this method were published and adopted, how quickly could a scam operator adapt to evade it, and what would that adaptation cost them?

Evading the phrase rules is trivial and immediate, since rewriting the signature paragraph defeats an exact match. Evading the archetype structure is harder, and the cost is the part worth discussing.

The Ghost Agency ran 99 postings across 60 job titles and 22 locations on one reused opening paragraph. That reuse is not laziness, it is the economics of the operation, since a single credible paragraph is amortized

across an entire campaign. Forcing genuine variation into every posting removes the economy of scale that makes volume fraud profitable, and volume is the business model.

The honest counterpoint, which I would rather raise myself than have raised for me, is that generative language models have made paraphrase close to free. That cost argument is considerably weaker in 2026 than it would have been in 2014, and it connects directly to my drift result. Part of what I measured may reflect a world in which posting text is increasingly machine generated, which would erode template based detection generally. That is a limitation on the future of the approach, not only on this dataset.

What resists adaptation is the thing an operator cannot fake cheaply. Both the Borrowed Brand and the Ghost Agency invested in appearing complete, and both would fail verification of employer identity immediately.

10. What would it cost, in compute and in human review time, to run this over a live job board processing thousands of postings a day?

Discovery and matching are different orders of magnitude and should be budgeted separately.

Discovery is the expensive half: embedding 42,327 text chunks, two dimensionality reductions and a density based clustering pass, which ran on a single mid range GPU. That is a periodic batch job, monthly or quarterly, not a per posting cost.

Matching is nearly free. Scoring a new posting is one TF-IDF transform and four dot products against the centroid matrix, on CPU, on the order of milliseconds. At ten thousand postings a day that is a rounding error on any infrastructure, and it requires shipping neither the embedding model, the reducer, nor the clustering model.

Human review is the real budget. At the union operating point, reviewing 9.8% of postings surfaces 71.7% of fraud. On ten thousand postings a day that is roughly 980 reviews. At two minutes per review, assuming a trained reviewer and no queue overhead, that is about 33 hours of review a day, or four to five full time reviewers. The tuning knob is the lift threshold: a board that can staff only two reviewers raises it and accepts lower recall, and the concentration measure exists so that trade can be made explicitly rather than by feel.

The number I would not promise is the 71.7%. Given the drift result, that figure has to be re-measured on current postings with refitted centroids before anyone plans headcount around it.